

Traditional Chinese Medicine Constitution Characteristics and Influencing Factors for Acute Mountain Sickness Susceptibility in Rapidly Ascending High-Altitude Populations: A Cross-Sectional Survey

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Abstract

Background: Acute Mountain Sickness(AMS) is a significant health concern for individuals rapidly ascending to high altitudes. In severe cases, the condition can develop into high-altitude pulmonary edema and high-altitude cerebral edema, which can have a significant impact on the quality of life of individuals entering Tibet and even pose a threat to their lives. The occurrence of AMS is closely related to factors such as altitude, the time of year of entry into Tibet, and individual activities. However, individual differences in the manifestation of AMS and the subsequent physiological responses are evident, with these responses varying among individuals of differing body types within the same environmental context. The Traditional Chinese Medical Constitution (TCMC) is a significant indicator of health status and may be associated with an increased propensity for plateau reactions. However, owing to a combination of environmental influences, subject demographics, and other factors, the current body of research related to AMS and TCMC is relatively limited.

Objective: To investigate AMS-susceptible TCMC types and key influencing factors in populations rapidly ascending to Tibet, providing evidence for TCM-based prevention and individualized AMS management.

Methods: This cross-sectional study enrolled 1,482 eligible participants (healthy males aged 18–40 years who were newly ascended to high altitude). Data collection included the following: 1) questionnaires (demographics, lifestyle, anxiety, insomnia); 2) physical examinations (vital signs: respiration, heart rate, SpO₂, blood pressure) and body mass index (BMI); 3) TCMC assessment via the *Classification and Determination of TCMC* standard; and 4) AMS diagnosis and severity grading via the Lake Louise Score (LLS). Spearman correlation was used to analyze the relationships between variables. Multivariate logistic regression (backward stepwise, entry $P < 0.25$) identified independent AMS risk factors.

Results: The overall incidence of AMS was 32.9% (488 out of 1482 cases). The prevalence of unbalanced constitutions was markedly greater in the AMS group than in the non-AMS group (32.6% vs. 7.6%; $\chi^2 = 161.8$, $P < 0.001$). Among patients with AMS and an unbalanced constitution, Qi-deficiency was the most prevalent, accounting for 78.0% of the cases. The incidence of AMS was observed to vary significantly according to different constitutions, with rates of 72.9% for Qi-deficiency and 26.3% for a Balanced constitution. Univariate analysis revealed significant correlations (all $P < 0.001$): SpO₂ was negatively correlated with AMS ($r = -0.231$); Insomnia ($r = 0.272$), Anxiety ($r = 0.297$), Unbalanced constitution tendency ($r = 0.333$), Season (winter = 1; $r = 0.122$), Altitude ($r = 0.157$), and BMI ($r = 0.127$) were positively correlated. Multivariate analysis revealed the following independent risk factors: Anxiety (aOR = 1.16, 95% CI: 1.08–1.25, $P < 0.001$), Insomnia (aOR = 1.09, 95% CI: 1.03–1.16, $P = 0.003$), Unbalanced constitution tendency (aOR = 1.06, 95% CI: 1.02–1.10, $P = 0.002$), and SpO₂ (aOR = 0.91, 95% CI: 0.87–0.95, $P < 0.001$). BMI showed a weak positive association (aOR = 1.02, 95% CI: 1.00–1.04; $P = 0.041$). Season, altitude, and blood pressure lost significance after adjustment ($P > 0.05$).

Conclusion: Qi-deficiency is the most common AMS-susceptible TCMC. Key modifiable risk factors include insomnia, anxiety, and low SpO₂. The influence of season and altitude are mediated through

physiological, psychological, and constitutional factors. These findings advocate for TCMC-differentiated prevention strategies for AMS. These strategies encompass preemptive qi-tonifying interventions for susceptible individuals, psychological support, and sleep optimization.

Introduction

AMS is a hypoxic reaction caused by rapid ascent to high altitudes. Its symptoms include headache, fatigue, insomnia, nausea, etc.^[1-3]. Severe cases can progress to high-altitude pulmonary edema and high-altitude cerebral edema, seriously affecting the quality of life of people entering Tibet and threatening their life and health.^[4-5] Globally, the incidence of AMS varies by region and population. Generally, above 2,500 m above sea level, the incidence rate can reach 20% to 30%. Even below 2,000 m above sea level, susceptible people may still develop AMS or even high-altitude pulmonary edema, which seriously affects the work and life of people entering Tibet^[6-8].

TCMC refers to the interactive relationships in aspects such as the morphology, physiological functions, and psychological states of an individual during the life process and is based on congenital endowment and acquired nurturing^[9]. TCMC are classified into balanced constitutions (normal constitutions) and unbalanced constitutions (biased constitutions, including Qi-deficiency constitution, Yang-deficiency constitution, Yin-deficiency constitution, Damp-heat constitution, Phlegm-dampness constitution, Blood stasis constitution, Qi stagnation constitution)^[10]. According to the theory of TCMC, the relationship between constitution and disease indicates that constitution determines disease susceptibility via gene–environment interactions^[11-14]. There are significant differences in the congenital endowment of an individual's TCMC, and the distribution of TCMC among different populations also shows remarkable disparities^[15-18]. Research evidence shows that constitutional imbalance is associated with the occurrence of diseases. At present, research on the relationship between TCMC and AMS is still in its infancy^[23-27]. It is restricted by regional and demographic factors and is also limited by an insufficient sample size, a lack of systematic experimental design, and long-term follow-up observations. A case–control study conducted by the Institute of Tibetan Medicine of the China Tibetology Research Center (n=113) revealed that the incidence of AMS was greater in individuals with the lung type of Tibetan constitution (corresponding to "qi deficiency" in TCM) than in individuals with other constitution types, with an odds ratio (OR) of 2.459, indicating that the risk of AMS in individuals with this constitution was approximately 2.4 times greater than that in the general population. The incidence of AMS was found to be greater in subjects with this particular body type, with an OR of 2.459. This finding indicates that the risk of AMS in individuals with this body type is approximately 2.4 times greater than that reported in the general population. However, owing to the limitation of the sample size, statistical significance was not achieved^[21].

Given that patients with AMS tend to exhibit population-specific predispositions, this paper explores the adaptation and disease susceptibility of individuals in high-altitude environments from the perspective of TCM. In this study, we successfully overcame the challenges of regional limitations, study participant

compliance, and study sample size. We identified the types of TCM susceptible to AMS and their causative factors through questionnaires and clinical observations. We also established an early warning program for TCM identification related to AMS in the population and developed and promoted TCM prevention and treatment programs suitable for people with different body types, giving full play to the strengths of traditional Chinese medicine to effectively improve the quality of life and health of the plateau population. Ultimately, we hope to provide a scientific basis for the “individualized” prevention and treatment of AMS on the basis of TCM theories and to provide new ideas for the study of plateau diseases.

Materials and Methods

Study Design and Participants

This cross-sectional study included 1,482 healthy male volunteers who rapidly ascended to Tibet (>3000 m) between December 2023 and July 2024. The inclusion criteria were as follows: 1) first ascent or rapid ascent from low (<1500 m) to high altitude (>3000 m); 2) aged 18–40 years; 3) ability to complete questionnaires; and 4) signed informed consent. The exclusion criteria were as follows: 1) major cardiopulmonary, hepatic, renal, endocrine, or neurological diseases; 2) acute infections within 2 weeks; 3) pregnancy/lactation (males only); and 4) medications affecting outcomes. Preascent medical screening excluded individuals with significant comorbidities.

Data collection and instruments

The survey was presented in the form of a structured questionnaire divided into three main parts.

Part 1: Sociodemographic Information

This part encompasses elements such as age, weight, height, body temperature, blood oxygen saturation, and blood pressure. These details serve as fundamental descriptors of the survey participants, providing a backdrop against which subsequent analyses can be set. Understanding the demographic makeup of a group is crucial, as it can potentially influence various aspects of the study outcomes, including the prevalence and manifestation of certain conditions.

Part 2: TCMC Scale

On the basis of the Classification and Judgment of TCMC standards, each item within this scale has an original minimum score of 1 point and a maximum score of 5 points. Nine subscales are calculated independently. Initially, the raw score of each subscale is computed by summing the scores of all individual items, i.e., Raw score = sum of the scores of each item. This raw score is subsequently converted into a transformed score, with the range of the transformed score for each subscale being from 1 to 100 points. The formula for this conversion is as follows: transformed score = $[(\text{raw score} - \text{number of items}) / (\text{number of items} \times 4)] \times 100$. The judgment criteria are as follows: when the transformed score of the balanced constitution is ≥ 60 points and the transformed scores of the other eight types of

unbalanced constitutions are all less than 30 points, the judgment is "yes". If the transformed score of the balanced constitution is >60 points and the transformed scores of the other eight unbalanced constitutions are all less than 40 points, it is judged as "basically yes". Otherwise, it is "no". For unbalanced constitutions, when the transformed score is ≥ 40 points, the judgment is "yes"; when it ranges from 30–39 points, it is "tend to be yes"; and when it is less than 30 points, it is "no". The higher the score of a subscale is, the more pronounced the tendency toward that particular constitution type. This section of the questionnaire delves into the concept of TCMC, which is integral to understanding an individual's predisposition to certain health conditions. By precisely defining and categorizing these constitutions, this study aims to explore potential relationships between constitution types and other variables under investigation^[28-31].

Part 3: LLS for Assessing the Severity of AMS

This section focuses on evaluating the severity of AMS through the LLS. It consists of several aspects: headache (scored from 0–3 points), nausea and vomiting (0–3 points), fatigue and weakness (0–3 points), insomnia (0–3 points), and other symptoms (0–3 points). A diagnosis of AMS is made when the total score is ≥ 3 points and accompanied by headache. Additionally, the AMS can be classified according to the score magnitude. A score of 3–5 points indicates mild AMS, a score of 6–9 points represents moderate AMS, and a score of 10–12 points signifies severe AMS. Measuring the severity of AMS is of utmost importance, as it not only helps in understanding the impact of the condition on the individual's health but also in formulating appropriate treatment and management strategies. This scoring system provides a standardized and objective way to quantify severity, enabling more accurate comparisons and analyses within a study^[32-34].

Quality Control

Standardized training, equipment, and protocols were used. The questionnaires underwent onsite verification. TCMC assessments were performed by certified TCM practitioners. Data entry utilized double verification.

Statistical analysis

The analyses were performed with SPSS 26.0. Continuous variables are presented as the means $\pm$ SDs (normally distributed) or medians (IQRs); categorical variables are presented as frequencies (%). Univariate analyses: Independent t test/Mann–Whitney U test (continuous), chi–square test/Fisher's exact test (categorical), Spearman correlation (continuous/ordinal vs. AMS). Variables with $P < 0.25$ or theoretical importance were included in multivariate binary logistic regression (backward stepwise LR, removal $P > 0.10$). The results are reported as adjusted Odds Ratios (aOR) with 95% Confidence Intervals (CI). Model fit was assessed via the Hosmer–Lemeshow test ($P > 0.05$) and Nagelkerke R^2 . Significance level: $\alpha = 0.05$ (two-tailed).

Results

Sociodemographic and physical characteristics

The subjects included healthy young adult males newly admitted to Tibet. Prior to their travel to Tibet, they underwent rigorous physical examinations that excluded the presence of major diseases. The statistical analysis incorporated survey data from a total of 1,482 subjects. The indicators of these subjects, including age, height, weight, respiration rate, heart rate, body temperature, blood oxygen saturation and blood pressure, are presented in Table 1.

Table 1. Baseline Demographic and Physiological Characteristics of Participants (n=1,482)

	Minimum	Maximum	Mean	Standard error
Age	18	39	21.61	.035
Height	154	197	173.87	.120
Weight	46	100	67.72	.171
BMI (Kg/m ²)	14.11	36.36	22.37	.047
Breath	10	32	17.92	.038
Heart rate	32	127	70.25	.190
Temperature	35	39	36.49	.005
SpO ₂	65	100	97.79	.031
Systolic blood pressure	78	173	111.53	.193
Diastolic blood pressure	37	121	64.82	.153

Body mass index (BMI) and vital signs of the subjects before and after entry into the plateau were compared. A significant increase in respiratory rate and heart rate was observed in subjects after they reached the plateau compared with before they reached the plateau ($P<0.001$). Concurrently, a significant decrease in oxygen saturation was observed ($P<0.001$), accompanied by substantial changes in systolic and diastolic blood pressure ($P<0.001$). However, the difference in body mass index remained insignificant ($P>0.05$). As shown in Figure 1.

AMS incidence

Among the subjects, 488 people were diagnosed with AMS, with an incidence rate of 32.9%. Specifically, the incidence rate of mild AMS was 28.5%, that of moderate AMS was 4.3%, and there were 2 cases of severe AMS, with an incidence rate of 0.1%, as shown in Figure 2A.

The differences in the incidence of AMS across different seasons and altitudes are shown in Figure 2B, 2C. The incidence of AMS is 42.0% in winter and 29.7% in summer. In addition, the symptoms of AMS are more severe in winter. Taking the altitude of 3000 m as the dividing line, the incidence of AMS in high-

elevation areas (above 3000 m) is 37.1%, whereas in low-elevation areas (below 3000 m), the incidence is 20.3%.

TCMC and AMS

Among the AMS population, 159 patients (32.6%) were diagnosed with unbalanced constitutions, whereas among the non-AMS population, 76 patients (7.6%) had unbalanced constitutions. The proportion of unbalanced constitutions in the AMS population was significantly greater than that in the non-AMS population, as shown in Figure 3A, 3B. Among the unbalanced constitutions, the Qi-deficiency constitution was the most common (78.0%), followed by the Yin-deficiency constitution (11.9%), the Yang-deficiency constitution (3.1%), and the Damp-heat constitution (2.5%). As shown in Figure 3C.

The incidence of AMS in different TCMC types was statistically analyzed. The incidence of AMS in the Qi-deficiency constitution was 72.9%, that in the Yang-deficiency constitution was 71.4%, that in the Yin-deficiency constitution was 55.9%, that in the Damp-heat constitution was 33.3%, and that in the balanced constitution was 26.3% (the constitutions of fewer than 10 people were not included in the calculation). As shown in Figure 3D.

AMS-Influencing Factors

Univariate analysis

The results of the correlation analysis revealed that SpO₂, Breath, Heart rate, Insomnia, Tension and anxiety, TCMC, Season, Altitude, and BMI significantly affected the occurrence of AMS. Negative correlation: SpO₂ (r=-0.231). Positive correlations: Respiration rate (r=0.110), Heart rate (r=0.128), Insomnia (r=0.272), Anxiety (r=0.297), Unbalanced constitution tendency (r=0.333), Season (winter=1; r=0.122), Altitude (r=0.157), and BMI (r=0.127). Blood pressure was weakly correlated (r<0.1, P<0.05). As shown in Figure 4A.

Multivariate logistic regression analysis

Variables with P<0.25 (SpO₂, Respiration, Heart rate, Insomnia, Anxiety, Unbalanced constitution tendency, Season, Altitude, BMI, SBP, and DBP) were included. The final model identified significant factors: Anxiety: aOR=1.16, 95% CI: 1.08–1.25, P<0.001; Insomnia: aOR=1.09, 95% CI: 1.03–1.16, P=0.003; Unbalanced constitution tendency: aOR=1.06, 95% CI: 1.02–1.10, P=0.002; SpO₂: aOR=0.91, 95% CI: 0.87–0.95, P<0.001; BMI: aOR=1.02, 95% CI: 1.00–1.04, P=0.041. Respiration rate, Heart rate, Season, Altitude, and Blood pressure were not significantly different (P>0.05). The model fit was good (Hosmer–Lemeshow χ^2 =7.24, P=0.510; Nagelkerke R²=0.286). As shown in Figure 4B and Table 2.

Table 2. Multivariate logistic regression analysis of factors influencing AMS (n=1,482)

Factors	<u>β-coefficient</u>	Std. Error	<u>aOR</u> (95% CI)	P-value
Anxiety	0.148	0.036	1.16 (1.08-1.25)	< 0.001
Insomnia	0.086	0.029	1.09(1.03-1.16)	0.003
Unbalanced Constitution	0.058	0.019	1.06 (1.02-1.10)	0.002
SpO ₂	-0.094	0.022	0.91 (0.87-0.95)	<0.001
BMI	0.020	0.010	1.02(1.00-1.04)	0.041
Constant	-0.752	0.350	0.47	0.032

Discussion

This large cross-sectional study identified Qi-deficiency as the predominant AMS-susceptible TCM constitution and confirmed anxiety, insomnia, and low SpO₂ as key modifiable risk factors in young males who rapidly ascended to Tibet.

In this study, the prevalence of AMS was 32.9%, which is consistent with reports from some previous regions^[4,5], confirming that AMS is a significant health problem faced by people rapidly ascending to high-altitude areas. The proportion of unbalanced constitutions in the AMS group was significantly greater than that in the non-AMS group (32.6% versus 7.6%), and Qi-deficiency constitution dominated (78.0%) among the unbalanced constitutions of AMS patients. According to Chinese medical theory, Qi is the fundamental substance that constitutes and maintains life in the human body. It performs various functions, such as promoting, warming, defending, consolidating and gasifying. If Qi is deficient, the functions of internal organs decline, and the body's ability to resist disease is weakened^[9,10]. In a low-oxygen plateau environment, the human body needs to mobilize more Qi to maintain normal physiological functions, such as respiration, blood circulation and energy metabolism. People with Qi-deficiency are more prone to headaches, fatigue, shortness of breath and other core symptoms of AMS due to insufficient and reduced Qi function, decreased adaptability to the low-oxygen plateau environment, poor Qi and blood circulation and obstructed Qi (oxygen) intake and distribution^[20,21]. This is highly consistent with basic TCM theory, which states that 'when positive Qi exists within, evil cannot interfere' and 'where evil comes from, its Qi must be deficient'. Our results are consistent with Luo et al.'s study of body mass and plateau adaptation^[23], and for the first time, in a large sample, we have clarified the central role of Qi-deficiency in AMS. This provides an important theoretical basis and practical target for 'preventing disease before it occurs' (e.g., taking Qi-tonifying herbs such as *Rhodiola rosea*, *Astragalus membranaceus* and *Radix et Rhizome Ginseng* in advance or using moxibustion on the qihai, guanyuan and zusanli acupuncture points).

Anxiety (aOR=1.16) and insomnia (aOR=1.09) were significant independent risk factors. TCM emphasizes the mind–body connection: anxiety disrupts Qi movement (*fright causes Qi disorder*)^[14],

whereas insomnia depletes Qi and Blood, impairing recovery (*Yang fails to enter Yin*)^[22]. Insomnia leads to “Yang not Yin”, Yin and Yang imbalance, Qi and Blood dark depletion. Under the low-oxygen environment of plateaus, the generation and operation of “zongqi” (closely related to cardiopulmonary function and oxygen utilization), which are already affected, are further impaired due to insomnia-induced loss of heart and spirit, deficiency of the liver and blood, so that the body's adaptive ability to the plateau environment is greatly reduced, thus inducing or aggravating AMS symptoms such as headache, vertigo, and extreme fatigue^[22]. In terms of emotion and will, “Suwen” clearly noted that “shock causes confusion of Qi” and that “thinking causes knotting of Qi”. Tension and anxiety, as strong negative emotional activities, can directly lead to disorganization of Qi and disorders of elevation at the inlet and outlet^[14]. Under the condition of insufficient clear Qi on the plateau, this Qi disorder further aggravates the imbalance of Qi, Blood, Yin and Yang in the body. Excessive anxiety and thoughts can easily damage the heart and spleen, and a deficiency of the heart and spleen results in a lack of sources of qi and blood biochemistry and weak operation^[9], which cannot effectively meet the extra demand for Qi and Blood under high-altitude stress, thus significantly increasing the risk of AMS and the severity of the disease. Modern medical research has confirmed that psychological stress and sleep deprivation can reduce the ability to adapt to altitude by activating the sympathetic-adrenomedullary system and the hypothalamic-pituitary-adrenocortical axis, exacerbating inflammatory responses and oxidative stress, and impairing cognitive function and physiological homeostasis^[5,26]. Therefore, in the prevention of AMS, in addition to focusing on physiological indicators (e.g., blood oxygen), positive psychological conditioning (e.g., relaxation training, positive thinking) and ensuring sufficient quality sleep are crucial.

SpO₂, the most direct physiological indicator of systemic oxygenation, was identified as a key protective factor against AMS in this study (aOR=0.91). This finding aligns fundamentally with the hypoxic pathology of AMS^[1-3]. From a Traditional Chinese Medicine perspective, a low SpO₂ reflects insufficient clear Qi and a decline in pectoral Qi, resulting in failure to nourish tissues. Maintaining higher SpO₂ signifies enhanced oxygen acquisition and utilization capacity—constituting a critical foundation for preserving vital Qi when resisting high-altitude hypoxia. This mechanistic understanding explains the efficacy of Western preventive approaches, including preacclimatization training (improving oxygen transport/utilization) and supplemental oxygen therapy when necessary^[4].

Notably, univariate analysis revealed significant associations of season (higher incidence in winter) and altitude with AMS, which aligns with the physical properties of environmental factors (harsher cold and dryness in winter, along with thinner air; more severe hypoxia at higher altitudes)^[4,6]. However, in the multivariate model, after adjusting for factors such as constitution, insomnia, anxiety, SpO₂, and BMI, the independent effects of season and altitude became nonsignificant. These findings suggest that the impacts of season and altitude on AMS may operate through mediating factors—such as physiological parameters (blood oxygen saturation), psychological states (anxiety), sleep quality, and stress responses among individuals with different constitutions—rather than through simple independent direct effects. In the final model, BMI was weakly positively correlated (aOR=1.02). The underlying mechanism may be related to obesity increasing metabolic oxygen consumption and impairing respiratory function.

The present study focused on young men to exclude the potential influence of other age groups and the hormonal changes that occur during women's menstrual cycles on TCMC and AMS. However, the study also has several limitations. First, the study population was limited to healthy men aged 18–40 years, which limits the generalizability of the results to women, elderly individuals and people with illnesses. Second, the cross-sectional design could not establish causality. Third, variables such as insomnia and anxiety were assessed mainly via self-reported scales and could be supplemented with objective measures (e.g., somatosensory and heart rate variability) in future studies. Fourth, long-term tracking of dynamic changes in physical fitness and AMS regression is lacking.

Future directions: 1) Include diverse populations (age, sex, comorbidities); 2) conduct prospective cohort studies; 3) investigate physiological/genetic markers of Qi deficiency in hypoxia; 4) test TCM interventions (herbs, acupuncture) for AMS prevention in RCTs; and 5) develop integrated AMS risk prediction models.

Conclusion

This study confirmed that among young males that rapidly ascended to the Tibetan Plateau, Qi-deficiency constitution was the most susceptible TCM body type to AMS. Insomnia, anxiety, and low blood oxygen saturation are key modifiable factors influencing the occurrence of AMS. The impact of seasonal variations and altitude on AMS may be mediated through physiological, psychological, and constitutional factors. These findings provide a crucial foundation for implementing constitution-based preventive strategies for AMS from the holistic TCM perspective of the ***Constitution-Environment-Mental State***. Preemptive TCM interventions (reinforcing Qi and consolidating the foundation) for individuals with Qi-deficiency; enhanced psychological adaptation and sleep management before entering high-altitude regions; and close monitoring of blood oxygen saturation. These integrated approaches may serve as effective strategies for reducing AMS risk. Future research should validate these conclusions in diverse populations and further explore the underlying biological mechanisms and translational applications.

Abbreviations

AMS: Acute mountain sickness

TCMC: Traditional Chinese Medicine Constitution

BMI: body mass index

LLS: Lake Louise Score

CI: Confidence Intervals

aOR adjusted Odds Ratios

SBP: Systolic Blood Pressure

DBP: Diastolic Blood Pressure

Declarations

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Authors' contributions

Mengyang Long—Conceptualization, Methodology, Writing-Original Draft. Bo Liu, Jia Liu, Han Li, Jing Lu, Yue Chen, Xing Gan, Zhengqi Pan, Chaowei Wang, Lin Wang, Pengcheng Zhu, Mengya Shan, Maoyuan Cheng—Data curation. Jiejun Yang—Investigation. Cuiyao Tang, Cai Jing, Wei Zheng—Formal analysis, Visualization. Ying Zhang—Writing-Review & Editing. Yunqiao Wang—Funding acquisition. Zihui Xu—Project administration, Supervision.

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Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Ethics approval and consent to participate

The study protocol and informed consent form were approved by the Ethics Committee of the Second Affiliated Hospital of Army Medical University, Chongqing, China (Approval No. Medical Ethics 2022-Research-052-03). Written informed consent was obtained from all participants.

Consent for publication

Not applicable.

Competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Figures

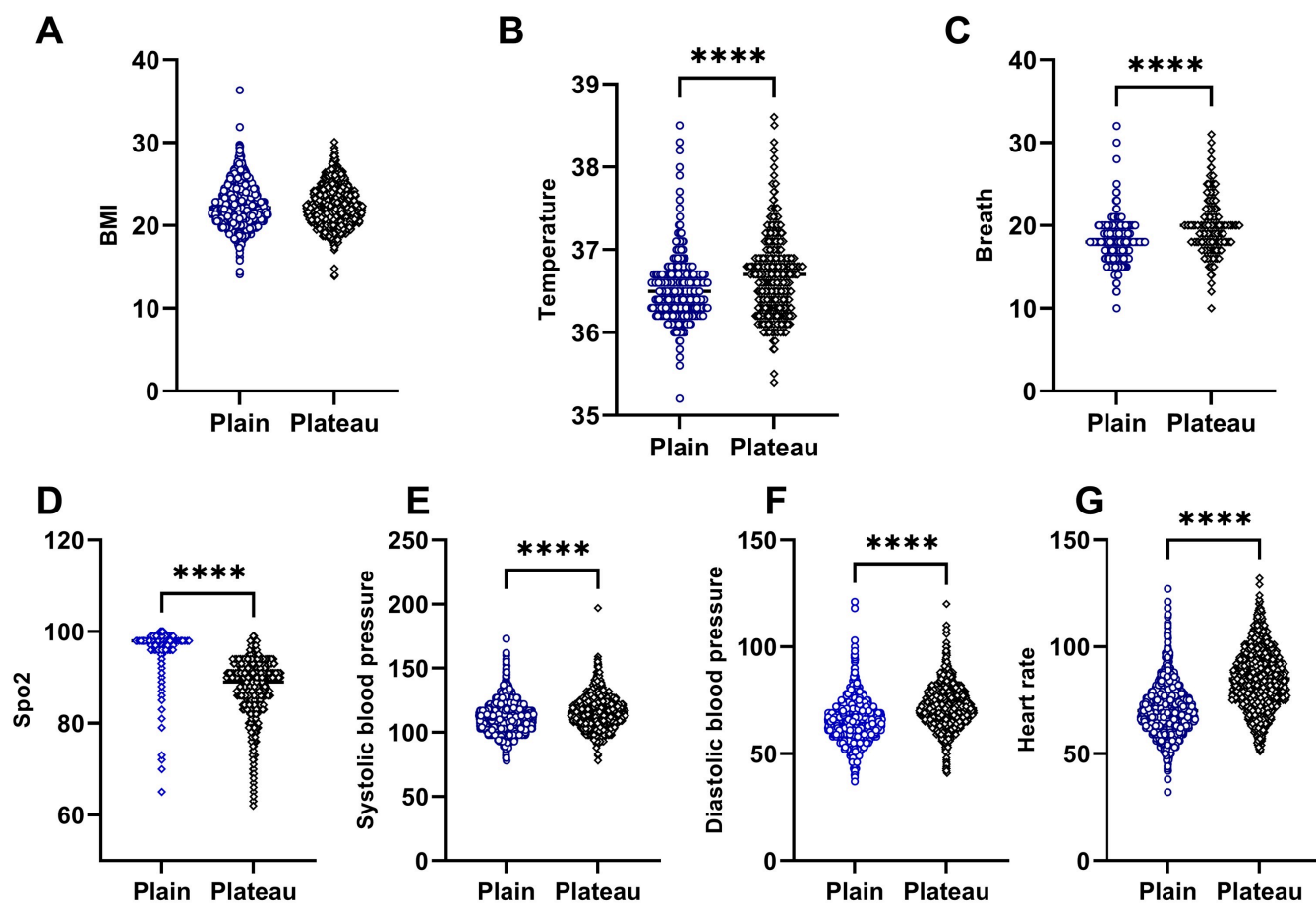


Figure 1

Comparison of vital signs before and after high-altitude warning(n=1,482)

A: BMI; B:Temperature; C:Breath; D:Spo2; E:Systolic blood pressure;
F: Diastolic blood pressure; G: Heart rate (****, p<0.0001)

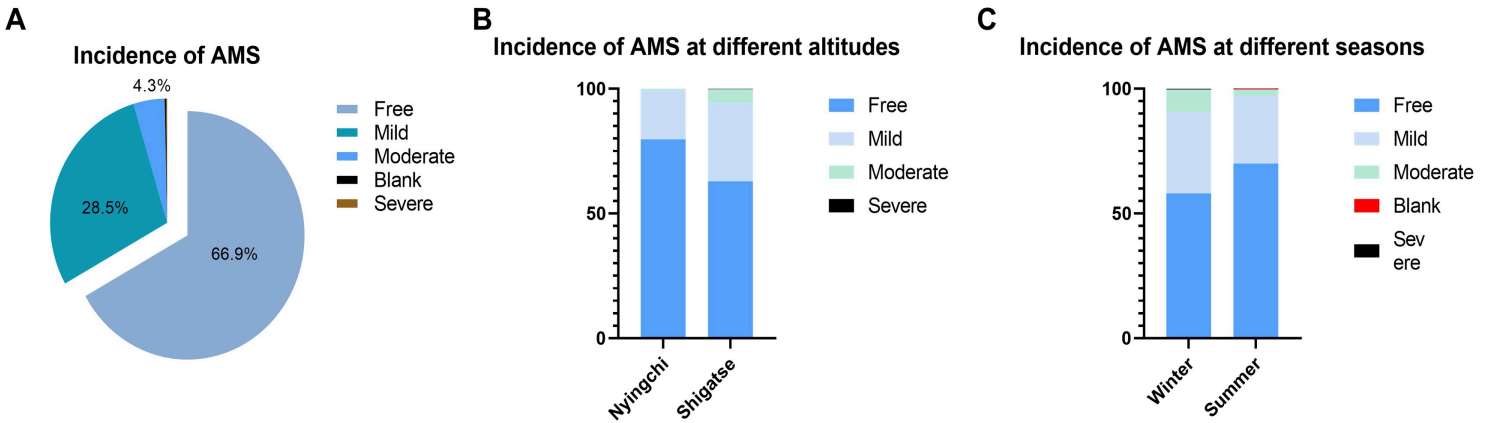


Figure 2

Incidence of AMS (n=1,482)

A: AMS incidence and severity distributions; B: AMS incidence by season; C: AMS incidence by altitude

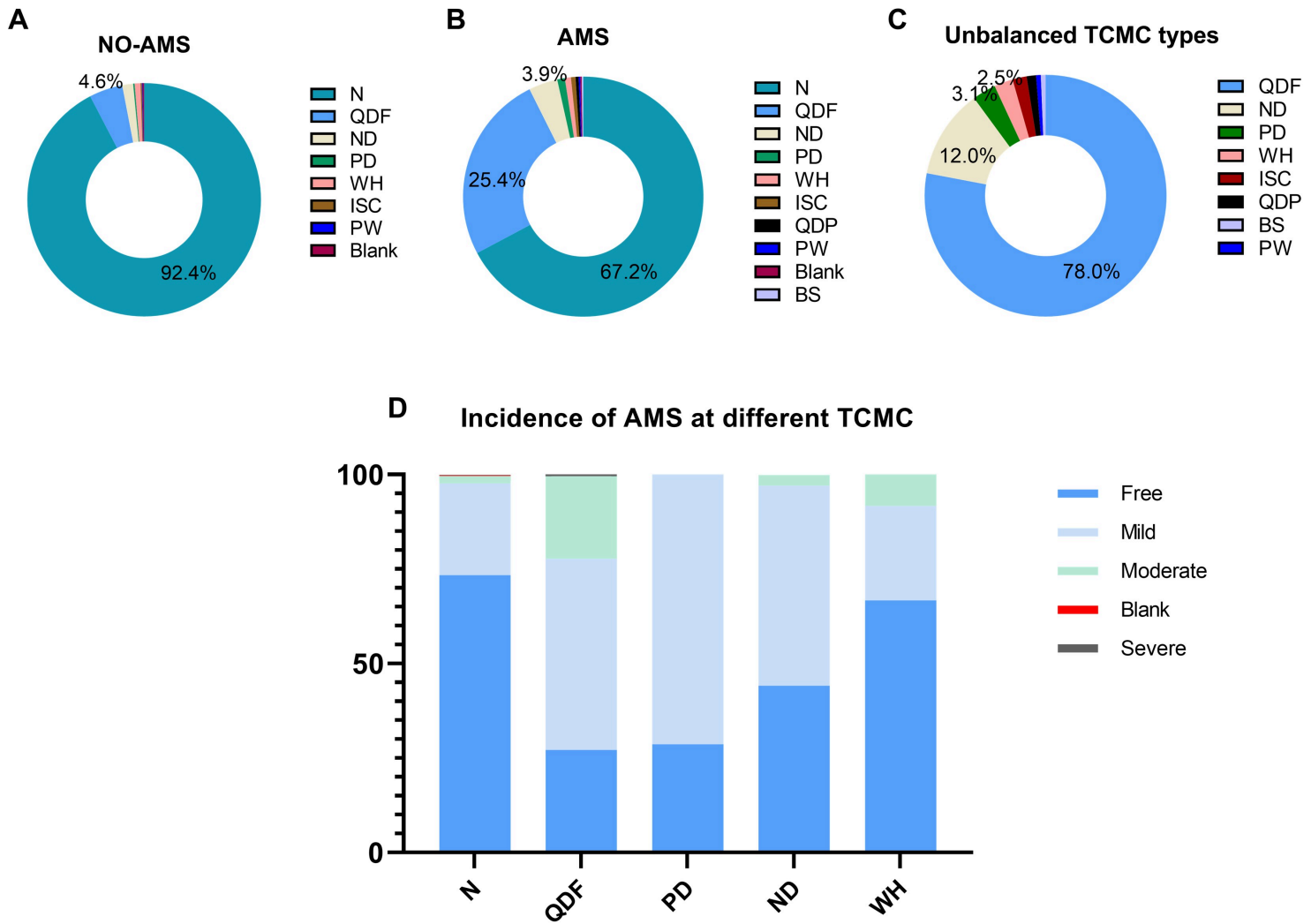


Figure 3

Distribution of TCM Constitutions (n=1,482)

A: Distribution of TCMC; B: Unbalanced TCMC types; C: Incidence of AMS in Different Constitutions
Abbreviations: N, Normality; QDF, Qi-deficiency; QDP, Qi-depression; PD, Yang-deficiency; ND, Yin-deficiency; PW, Phlegm-wetness; WH, Wetness-heat; BS, Blood stasis; ISC, Inherited special constitution; Blank, No corresponding constitution.

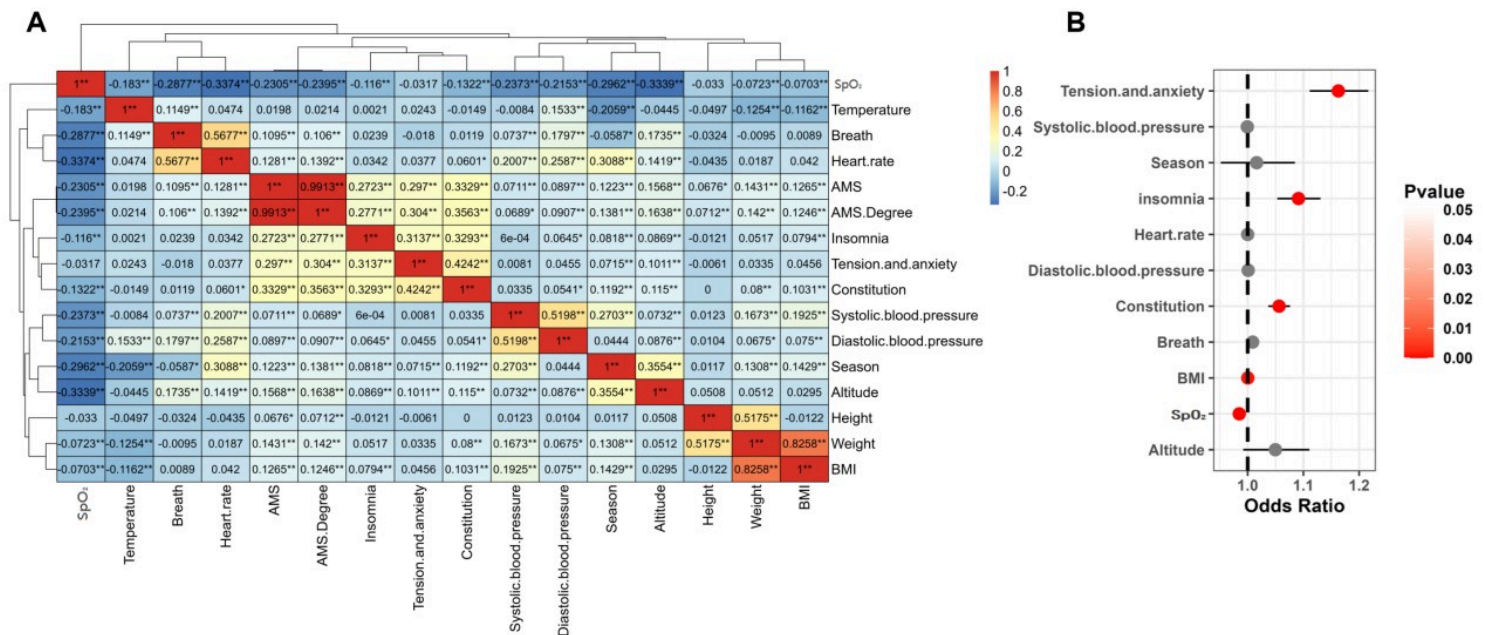


Figure 4

Factors potentially related to the incidence of AMS (n=1,482)

A: Univariate correlation analysis of factors associated with AMS onset; B: Forest plot of independent risk factors for AMS